# Xingjian Diao

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Linkedin | Github | Google Scholar

## Research Interests

My research focuses on **post-training** and **reinforcement learning** for **MLLMs** and **agents**. I train LLMs and MLLMs with **synthetic data flywheels**, multi-objective rewards, and fine-grained credit assignment, targeting reasoning and long-horizon decision-making in GUI, tool-use, and multimodal agentic settings. I also design **memory** mechanisms for long-context and long-horizon reasoning. My GitHub repositories have received 1.5k+ stars ⭐.

## Education

- **Dartmouth College** — *Sep 2022 - Mar 2027 (Expected)*
  *Ph.D. candidate in Computer Science* — Hanover, NH, USA
  - **Advisor**: Prof. Soroush Vosoughi and Prof. Jiang Gui
- **Northwestern University** — *Sep 2020 - Dec 2021*
  *Master of Science, Computer Science* — Evanston, IL, USA
  - **Advisor**: Prof. Nabil Alshurafa
- **University of Pittsburgh** — *Aug 2016 - Apr 2020*
  *Bachelor of Science, Computer Science* — Pittsburgh, PA, USA

## Industry Experience

- **Amazon** — *Jun 2026 - Sep 2026*
  *Applied Scientist Intern @ RBKS AI (Full-Time Return Inclined)* — Sunnyvale, CA, USA

  **A Verifiable Data Flywheel for Mobile GUI Agents**

  Pdf | Built TRAVEC, an end-to-end post-training framework for mobile GUI agents spanning data generation, supervised training, self-generated environment rollouts, offline RL, and large-scale evaluation. Created 550K verifiable training examples from heterogeneous GUI datasets, including evidence-grounded transition contracts, hard negatives, recovery trajectories, preference pairs, and verifier-calibration data. Trained specialized Action, Contract, and Verifier policies and designed transition- and recovery-aware GRPO rewards to distill verification and failure recovery into the final action policy. Starting from Qwen3.5-9B, TRAVEC outperforms seven frontier GUI agents across six static and interactive benchmarks while running 3.6–16.4× faster per action with a single action-only model call and no auxiliary reasoning or verification at inference.

  Synthetic Data Flywheel | GUI Agents | Post-Training

- **Samsung Research America** — *Mar 2026 - Jun 2026*
  *NLP Research Intern @ Language Intelligence Team, AI Center* — Mountain View, CA, USA

  **Doc-to-Atom: Learning to Compile and Compose Memory Atoms**

  Pdf | Built Doc-to-Atom, a compositional parametric-memory framework that decomposes documents into semantically typed knowledge atoms and compiles each atom into a micro-LoRA adapter, an optional micro-KV prototype, and a retrieval key via a hypernetwork. A two-stage router dynamically composes only query-relevant atoms into a query-specific adapter at inference. Trained end-to-end with a multi-objective distillation curriculum, it outperforms document-to-LoRA baselines across six diverse QA benchmarks while reducing compile-time GPU memory by up to 85%.

  Synthetic Supervision | Parametric Memory | Context Distillation

- **Amazon** — *Jun 2025 - Sep 2025*
  *Applied Scientist Intern @ Last Mile (Full-Time Return Inclined)* — Santa Cruz, CA, USA

  **High-Frequency Video-to-IMU Synthesis via Physics-Guided Simulation and Hybrid U-Net Refinement**

  Pdf | Proposed PrimeIMU, a physics-guided video-to-IMU generation framework that fuses low-frequency kinematic cues from 3D video poses with physics-inspired simulated inertial initialization through a hybrid U-Net refinement module, effectively bridging the anatomical–inertial gap to produce high-fidelity, sensor-faithful IMU signals that generalize across activities, devices, and datasets, enabling synthetic-only training, cross-domain adaptation, and scalable deployment of wearable sensing models.

  Synthetic Supervision | Ubiquitous Computing

  **IMU2Reason: A Multimodal LLM for Safety-Aware Activity Understanding**

  Pdf | Proposed MHIA-LM, a multimodal LLM trained on fused video and IMU representations, where spatiotemporal video embeddings from InternVideo2 and IMU motion features encoded by a PatchTST-based encoder are integrated through a multi-layer Gated Hierarchical Interaction module and mapped into the LLM token embedding space for instruction tuning, enabling fine-grained activity recognition and safety reasoning.

  MLLM | Multimodal Reasoning

## SELECTED 1ST-AUTHOR PUBLICATIONS [FULL LIST]

**[Preprint 2026] A Verifiable Data Flywheel for Mobile GUI Agents**
**Xingjian Diao** *et al.*
Pdf | Built TRAVEC, an end-to-end post-training framework for mobile GUI agents spanning data generation, SFT, environment rollouts, offline RL, and evaluation. Created 550K verifiable training examples with transition contracts, hard negatives, recovery trajectories, preference pairs, and verifier-calibration data, then trained Action, Contract, and Verifier policies with transition- and recovery-aware GRPO rewards. Starting from Qwen3.5-9B, TRAVEC outperforms seven frontier GUI agents across six benchmarks while running 3.6–16.4× faster per action with a single inference-time model call.
Synthetic Data Flywheel GUI Agents Post-Training

**[Preprint 2026] Of Errors and Echoes: Responsibility-Aware Credit Assignment for Tool-Use Agents**
**Xingjian Diao***, Tianyu Yang*, Wenjun Huang, Xiangchi Yuan, Chunhui Zhang, Xingchen Zhao, Weiyi Wu, Haotian Xu, Soroush Vosoughi, Xiangliang Zhang, Jiang Gui
Pdf | Developed Error-Chain-aware Optimization (ECHO), a fine-grained credit assignment framework for long-horizon tool-use agents that distinguishes error sources, propagated consequences, missed corrections, and recoveries. ECHO combines frozen-judge responsibility assessment with shared-consequence accounting and source-funded recovery to produce action-level training signals, improving performance across six tool-use benchmarks at 4B and 8B scales.
Tool-Use Agents Post-Training Credit Assignment

**[ACL 2026] Addressing Overthinking in Large Vision-Language Models via Gated Perception-Reasoning Optimization**
**Xingjian Diao**, Zheyuan Liu, Chunhui Zhang, Weiyi Wu, Keyi Kong, Lin Shi, Kaize Ding, Soroush Vosoughi, Jiang Gui
Pdf | Introduced Gated Perception–Reasoning Optimization (GPRO), a token-level adaptive computation framework that leverages large-scale perception–reasoning failure attribution (≈790K samples) to train a meta-reasoning controller via multi-objective reinforcement learning, dynamically routing between fast execution, visual re-perception, and slow reasoning to mitigate overthinking while improving both accuracy and efficiency in vision–language models.
Synthetic Supervision MLLM Post-Training

**[Preprint 2026] Doc-to-Atom: Learning to Compile and Compose Memory Atoms**
**Xingjian Diao**, Wenbo Li, Yashas Malur Saidutta, Avinash Amballa, Lazar Valkov, Srinivas Chappidi
Pdf | Proposed Doc-to-Atom, a compositional parametric-memory framework that decomposes each document into semantically typed knowledge atoms, compiles them via a hypernetwork into per-atom micro-LoRA adapters with retrieval keys, and trains a two-stage router to assemble only the query-relevant atoms into a query-specific adapter, outperforming document-to-LoRA baselines across six QA benchmarks at up to 85% less GPU memory.
Synthetic Supervision Parametric Memory Context Distillation

**[EMNLP 2025] SoundMind: RL-Incentivized Logic Reasoning for Audio-Language Models**
**Oral Presentation Award (top 4.35%)**, 30th EMNLP
**Xingjian Diao**, Chunhui Zhang, Keyi Kong, Weiyi Wu, Chiyu Ma, Zhongyu Ouyang, Peijun Qing, Soroush Vosoughi, Jiang Gui
Pdf | Github Code; ⭐ **Starred 1k+** | Dataset | Introduced the Audio Logical Reasoning (ALR) task containing 6,446 text–audio CoT-annotated samples to enable complex reasoning over spoken content, and proposed SoundMind, a rule-based reinforcement learning algorithm that enhances deep cross-modal reasoning in audio-language models.
Synthetic Supervision MLLM Post-Training

**[EMNLP 2025] ProtoVQA: An Adaptable Prototypical Framework for Explainable Fine-Grained Visual Question Answering**
**Oral Presentation Award (top 4.35%)**, 30th EMNLP
**Xingjian Diao***, Weiyi Wu*, Keyi Kong, Peijun Qing, Ming Cheng, Soroush Vosoughi, Jiang Gui
Pdf | Proposed ProtoVQA, an adaptable prototypical VQA framework that learns question-aware prototypes and uses spatially-constrained greedy matching to ground answers in semantically coherent image regions, unifying answering and grounding via a shared backbone and introducing the VLAS metric to quantify visual–linguistic alignment.
Multimodal QA Interpretability

**[NAACL 2025] Temporal Working Memory: Query-Guided Temporal Segment Refinement for Enhanced Multimodal Understanding**
**Xingjian Diao***, Chunhui Zhang*, Weiyi Wu, Zhongyu Ouyang, Peijun Qing, Ming Cheng, Soroush Vosoughi, Jiang Gui
Pdf | Github Code; ⭐ **Starred 300+** | Proposed Temporal Working Memory, a plug-and-play query-guided segment refinement module that maintains dynamic temporal memory to effectively preserve task-relevant video–audio segment and enhance long-range temporal reasoning, achieving consistent performance gains when integrated into nine recent state-of-the-art multimodal large language models (MLLMs) across AVQA, video captioning, and retrieval tasks.
MLLM Video Understanding

**[EMNLP 2024] Learning Musical Representations for Music Performance Question Answering**
**Xingjian Diao**, Chunhui Zhang, Tingxuan Wu, Ming Cheng, Zhongyu Ouyang, Weiyi Wu, Jiang Gui
Pdf | Proposed a specialized framework for audio–visual modeling in music understanding, addressing underexplored multimodal interactions, distinctive musical characteristics, and temporal alignment, and introduced annotated rhythmic

and source features, with the framework achieving state-of-the-art on Music-AVQA 1.0 and 2.0.
Multimodal QA Representation Learning

**[ACL 2025] Learning Sparsity for Effective and Efficient Music Performance Question Answering**
**Xingjian Diao**, Tianzhen Yang, Chunhui Zhang, Weiyi Wu, Ming Cheng, Jiang Gui
Pdf | Proposed Sparsify, a sparse learning framework for Music Audio-Visual Question Answering that integrates Sparse Masking, Adaptive Sparse Merging, and Sparse Subset Selection to reduce multimodal redundancy, highlight task-critical tokens, and accelerate training convergence, achieving efficiency and accuracy gains across Music-AVQA benchmarks.
Multimodal QA Sparsity Learning

**[ACL 2026] Music Performance Audio-Visual Question Answering Requires Specialized Multimodal Designs**
Wenhao You*, **Xingjian Diao***, Wenjun Huang, Chunhui Zhang, Keyi Kong, Weiyi Wu, Chiyu Ma, Zhongyu Ouyang, Tingxuan Wu, Ming Cheng, Soroush Vosoughi, Jiang Gui
Pdf | Providing a systematic analysis of how specialized multimodal architectures with spatio-temporal modeling enable reliable reasoning over musical performances.
Multimodal QA Video Understanding

**[WACV 2025] FT2TF: First-Person Statement Text-To-Talking Face Generation**
**Xingjian Diao**, Ming Cheng, Wayner Barrios, SouYoung Jin
Pdf | Proposed and developed a one-stage end-to-end text-to-talking face generation pipeline driven by first-person statement text, requiring only visual and textual inputs during inference. Experiments on LRS2 and LRS3 demonstrate state-of-the-art performance, showing its ability to generate realistic talking faces effectively from text inputs.
AIGC Multimodal Alignment

## Selected Collaborative Publications [Full List]

**[Preprint 2026] Primary Advantage Policy Optimization**
Xiangchi Yuan, **Xingjian Diao** *et al.*
Pdf | Developed Primary Advantage Policy Optimization (PAPO) for multi-objective LLM post-training, introducing asymmetric advantage fusion that preserves the primary update direction while retaining useful auxiliary supervision. Implemented sequence-level fusion for GRPO and token-level fusion for PPO, with independently normalized objectives and a dual-head critic for objective-specific GAE and return estimation. Validated across reasoning, tool-use, and multimodal safety tasks, consistently improving primary performance without sacrificing auxiliary objectives.
Multi-Objective RL Post-Training

**[Preprint 2026] Linear Attention Scales Better with Shared Latent Memory Projection**
Chunhui Zhang, **Xingjian Diao**, Xiangchi Yuan, Chiyu Ma, Yaning Jia, Zhongyu Ouyang, Soroush Vosoughi
Pdf | Proposed Mixture-of-Latent-Memories (MoLM), a scalable multi-memory linear-attention architecture that compresses each token into a shared latent representation and decodes it into memory-specific key-value states through lightweight per-memory projections, decoupling memory count from projection cost and enabling higher throughput, lower GPU memory usage, and stronger long-context performance than Mixture-of-Memories baselines.
Linear Attention Mixture-of-Memories

**[ACL 2026] What Makes a Good Curriculum? Disentangling the Effects of Data Ordering on LLM Mathematical Reasoning**
Yaning Jia, Chunhui Zhang, **Xingjian Diao**, Xiangchi Yuan, Zhongyu Ouyang, Chiyu Ma, Soroush Vosoughi
Pdf | Proposed a unified offline curriculum learning framework that systematically disentangles five curriculum dimensions and isolates the causal impact of data ordering to provide principled guidance on curriculum design.
LLM Curriculum Learning

**[EACL 2026] Tailoring Memory Granularity for Multi-Hop Reasoning over Long Contexts**
Peijun Qing, **Xingjian Diao**, Chiyu Ma, Saeed Hassanpour, Soroush Vosoughi
Pdf | Proposed Tailoring Memory Granularity, a reward-guided framework that adaptively composes hybrid memory across multiple granularities, including chunks, triples, atomic facts, and summaries, to enhance large language models in long-context multi-hop reasoning through dynamic query-specific memory selection.
LLM Hybrid Memory Multi-hop Reasoning

**[EMNLP 2026] Which Tokens Should SFT Actually Learn? A Token-Trimming Perspective on Mathematical Reasoning**
Yaning Jia, Chunhui Zhang, Wenxuan Xu, **Xingjian Diao**, Xiaoyuan Wang, Soroush Vosoughi
Pdf | Proposed Trimmed Logit-Gap SFT (TrimSFT), a token-level reweighting framework that selectively concentrates supervision on intermediate-confidence tokens by weighting the SFT loss according to the logit gap between the gold token and its strongest competitor, trimming both already-mastered and weakly supported tokens to enhance mathematical reasoning in large language models without additional models or forward passes.
Post-Training Token Reweighting

**[CVPR 2026] Learning Spatial-Preserving Hierarchical Representations for Digital Pathology**
Weiyi Wu, **Xingjian Diao**, Chunhui Zhang, Chongyang Gao, Xinwen Xu, Siting Li, Jiang Gui
Pdf | Introduced Sparse Pyramid Attention Networks (SPAN) for gigapixel whole-slide pathology, combining spatial-adaptive condensation with context-aware refinement to preserve spatial structure and enable efficient multi-scale

tumor detection, classification, and segmentation.
Digital Pathology Whole Slide Image Analysis

**[ICML 2026] HiST: A Hierarchical Sparse Transformer for Cross-Modal Spatial Transcriptomics Modeling**
Weiyi Wu, Xinwen Xu, **Xingjian Diao**, Siting Li, Zhi Wei, Alma Andersson, Jiang Gui
Pdf | Proposed HiST, a hierarchical sparse transformer for H&E-to-ST inference that models measured spots as a lattice-indexed sparse field to capture multiscale tissue context efficiently, improving predictive performance while substantially reducing inference time and memory across multi-organ spatial transcriptomics benchmarks.
Spatial Transcriptomics

**[Preprint 2026] ADAPT: Audio-Depth Action Planning with Transition-Aware Rewards for Lights-Out Industrial Navigation**
Wenjun Huang, Shenghao Fu, Yian Jin, Yang Ni, Yifan Li, Lichi Li, **Xingjian Diao**, Pengfei Zhang, Yutong Song, Hanning Chen, Jiarui Cui, Tianyu Yang, Shilong Li, Yu Kong, Mohsen Imani
Pdf | Developed Audio-Depth Action Planning (ADAPT), a multimodal framework for RGB-free industrial navigation integrating spatial audio, depth perception, and reinforcement learning. Introduced DarkNav-Bench with 24 industrial scenes and Transition-Aware Reward Shaping (TARS), which uses a pose-aware shortest-action potential to distinguish beneficial rotations from counterproductive actions. ADAPT combines multimodal SFT with PPO-based closed-loop optimization, outperforming navigation baselines across seen and unseen environments.
Embodied AI MLLM Reinforcement Learning

**[CIKM 2026] The Role of User-Item Connectivity in CTR Prediction**
Zhongyu Ouyang, Chunhui Zhang, **Xingjian Diao**, Yijun Tian and Soroush Vosoughi
Pdf | Studied the role of user–item connectivity in Click-through rate (CTR) prediction, using a minimal message-passing operator as a controlled probe to isolate graph-connectivity effects from CTR model architecture. The work shows that connectivity smooths user/item representations, improves stability under sufficient graph density, and can be efficiently integrated through fine-tuning while redistributing predictive gains toward under-served user groups.
CTR Prediction Message Passing Recommender Systems

**[TMLR 2026] The Pitfalls of Text Degeneration when Aligning LLMs through Model Merge**
Peijun Qing, Hefan Zhang, Lei Hsiung, Haiquan Lu, **Xingjian Diao**, Chiyu Ma, Saeed Hassanpour, Soroush Vosoughi
Pdf | Systematically investigated how alignment-oriented model merging in decoder-based LLMs can induce localized text degeneration, including repetition, unintended code switching, and nonsensical outputs, even when standard reward and performance metrics improve, through a large scale analysis of extrapolation and interpolation merges across multiple models and merging methods.
Model Merge LLM

**[WACV 2026] Exploiting Label-Independent Regularization from Spatial Patterns for Whole Slide Image Analysis**
Weiyi Wu, Xinwen Xu, Chongyang Gao, **Xingjian Diao**, Siting Li, Jiang Gui
Pdf | Proposed SRMIL for whole-slide image analysis, combining label-guided classification with label-independent spatial reconstruction to exploit spatial patterns as regularization, improve feature learning, and enhance robust weakly supervised classification.
Digital Pathology

**[AACL 2025] Judging the Judges: A Systematic Study of Position Bias in LLM-as-a-Judge**
**Oral Presentation Award (top 3.30%)**, 14th IJCNLP-AACL
Lin Shi, Chiyu Ma, Wenhua Liang, **Xingjian Diao**, Weicheng Ma, Soroush Vosoughi
Pdf | Proposed a systematic position-bias evaluation framework for LLM-as-a-Judge that integrates Repetition Stability, Position Consistency, and Preference Fairness to measure bias robustness, expose primacy–recency behaviors under prompt permutations, and extend analysis from pairwise to list-wise comparisons on MTBench and DevBench.
LLM-as-a-Judge

**[EMNLP 2025] Knowing More, Acting Better: Hierarchical Representation for Embodied Decision-Making**
Chunhui Zhang, Zhongyu Ouyang, **Xingjian Diao**, Zheyuan Liu, Soroush Vosoughi
Pdf | Proposed a hierarchical action probing framework that aggregates layer-wise MLLM representations to enhance spatial grounding, contextual integration, and abstract reasoning for embodied decision-making, achieving substantial gains across language-guided rearrangement tasks.
MLLM Vision–Language–Action

**[EMNLP 2025] Assessing and Mitigating Medical Knowledge Drift and Conflicts in Large Language Models**
Weiyi Wu, Xinwen Xu, Chongyang Gao, **Xingjian Diao**, Siting Li, Lucas A Salas, Jiang Gui
Pdf | Proposed ConflictMedQA, a guideline-grounded benchmark pairing up-to-date and outdated clinical recommendations, along with ECDA/IKCR metrics to evaluate medical knowledge drift and assess RAG, DPO, and RoD for resolving temporal conflicts in LLMs.
Medical LLM Knowledge Drift RAG

**[Preprint 2025] On The Design Choices of Next Level LLMs**
Yijun Tian, **Xingjian Diao**, Ming Cheng, Chunhui Zhang, Jiang Gui, Soroush Vosoughi, Xiangliang Zhang, Nitesh V. Chawla, Shichao Pei
Pdf | Provided a comprehensive analysis of current LLM design choices across model architecture, attention mechanisms,

post-training strategies, optimization techniques, and data selection, identifying key trends and proposing future research directions for next-generation large language models.
LLM Post-Training Reinforcement Learning

**[EMNLP 2024] AlphaLoRA: Assigning LoRA Experts Based on Layer Training Quality**
Peijun Qing, Chongyang Gao, Yefan Zhou, **Xingjian Diao**, Yaoqing Yang, Soroush Vosoughi
Pdf | Proposed AlphaLoRA, a training-free layer-wise expert allocation strategy for LoRA-MoE that leverages Heavy-Tailed Self-Regularization to quantify layer training quality and assign experts accordingly, reducing redundancy while improving performance across NLP and reasoning benchmarks.
LLM Mixture-of-Experts

**[EMBC 2024] GluMarker: A Novel Predictive Modeling of Glycemic Control Through Digital Biomarkers**
**EMBC NextGen Scholar Award**, IEEE 46$^{th}$ EMBC
Ziyi Zhou*, Ming Cheng*, **Xingjian Diao***, Yanjun Cui, Xiangling Li
Pdf | Proposed GluMarker, a digital biomarker framework that integrates broader daily factors (meals, insulin doses, and CGM-derived metrics) to predict next-day glycemic control and identify key digital biomarkers that reveal how everyday behaviors shape diabetes management.
Digital Biomarkers

## Teaching

**Graduate Teaching Assistant**

- COSC89/189 (Video Understanding), Dartmouth College, Spring 2024
- COSC74/274 (Machine Learning), Dartmouth College, Winter 2024
- COSC61 (Database Systems), Dartmouth College, Summer 2023
- COSC10 (Object Oriented Programming), Dartmouth College, Spring 2023
- COSC62/162 (Applied Cryptography), Dartmouth College, Winter 2023
- COSC10 (Object Oriented Programming), Dartmouth College, Fall 2022

## Service

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|---|---|
| **Area / Session Chair:** | ACL 2026, ICASSP 2027, NeurIPS VLM4RWD Workshop 2026 |
| **Reviewer / PC:** | ICML 2026, NeurIPS 2025–2026, ICLR 2025–2027, CVPR 2025–2026, ICCV 2025, ACL 2025–2026, ACL Industry Track 2025–2026, EMNLP 2025–2026, NAACL 2027, AAAI 2027, ACMMM 2025–2026, AISTATS 2026, EACL 2026–2027, WACV 2025, ICASSP 2025–2026, IUI 2026, ISBI 2025–2026, BMVC 2026, IJCNN 2024–2026, ICME 2024–2026, BHI 2026, AVSS 2025 |
| **Journal Reviewer:** | TMLR 2025–2026, IEEE TAI 2026, IEEE TAFFC 2026, IEEE JBHI 2026, FITEE 2026 |
| **Workshop Reviewer:** | Berlin Machine Learning Workshop 2026, NeurIPS MINT Workshop 2024 |

## Awards

- **Dartmouth Fellowship,** Dartmouth College. *2022-Present*
- **Biomedical Data Science Award,** Dartmouth College. *2026*
- **EMNLP Oral Presentation Award (top 4.35%),** 30$^{th}$ EMNLP. *2025*
- **IJCNLP-AACL Oral Presentation Award (top 3.30%),** 14$^{th}$ IJCNLP-AACL. *2025*
- **Guarini School of Graduate and Advanced Studies Award,** Dartmouth College. *2025*
- **Biomedical Data Science Award,** Dartmouth College. *2025*
- **IEEE EMBC NextGen Scholar Award,** IEEE EMBC. *2024*